REFERENCE · CHAPTERS 10 & 11

Private Service Connect (3 Doors) & VPC Service Controls Data Perimeter

Complete private networking and data-perimeter topology connecting managed agent runtimes to private MCP servers and databases.

1. The Three Private Service Connect (PSC) Constructs

Instead of legacy VPC Network Peering—which exposes entire L3 CIDR ranges, fails on overlapping IPs, and is non-transitive—our architecture uses three one-way Private Service Connect doors inside a **VPC Service Controls (VPC-SC)** perimeter:

PSC CONSTRUCT	DIRECTION	SUBNET / IP	ROLE IN THE ARCHITECTURE
1. PSC Network Attachment (`PSC interface`)	Google-Managed Agent / Egress Gateway → Your VPC	`10.10.9.0/28` (e.g. `10.10.9.2`)	Drops a virtual network card into your VPC so outbound traffic from the managed Egress Gateway enters your VPC with a predictable local source IP.
2. PSC Endpoint for Google APIs (`vpc-sc` bundle)	Your VPC → Internal Cloud Run / BigQuery / Spanner	`10.255.0.10` + Private Cloud DNS (`*.run.app` , `*.googleapis.com`)	Routes calls from the PSC interface to internal Cloud Run (`MCP Toolbox`) and Google APIs strictly inside the VPC-SC perimeter without public routing.
3. PSC Endpoint for Database Producer	Your VPC (`tools-subnet`) → Database Service Attachment	`10.10.2.50:5432`	Connects `MCP Toolbox` to the managed database instance over a one-way Andromeda NAT tunnel with zero public IP and zero VPC peering.